ABSTRACT

Radiation monitoring and mapping play a crucial role in nuclear safety, environmental surveillance, and emergency response systems. This Summer Internship Report presents the research and development work carried out on the project **"RADMAP – RADiation MAPing – GPS Enabled Radiation Mapping Prototype on Raspberry Pi-5 (3D Visualization)"** at the **Amity Institute of Nuclear Science & Technology (AINST)**. The primary objective of the project was to develop a portable and intelligent radiation mapping system capable of acquiring real-time radiation measurements along with geographical coordinates, environmental parameters, and advanced visualization for effective radiation monitoring and future source localization.

Throughout the **47-day internship**, extensive research and development activities were performed in the areas of embedded systems, geospatial mapping, radiation detection, and data visualization. The work involved developing and preprocessing a radiation dataset, publishing the dataset on Kaggle, and utilizing it for generating interactive **2D radiation heatmaps**. A real-time GPS acquisition module was developed using the **GlobalSat BU-353N GPS Receiver** to obtain latitude, longitude, UTC time, temperature, and humidity data, which were integrated with **Raspberry Pi-5** for continuous environmental monitoring and data processing.

The project further involved the integration and validation of a **CNSPEC Multi-Channel Analyzer (Gamma Spectrometer)** using standard radioactive sources such as **Cobalt-60 (Co-60)** and **Cesium (Cs-137)**. Radiation count rates, spectral peaks, and detector responses were successfully analyzed to generate dynamic radiation intensity maps. To overcome the limitation of conventional 2D heatmaps, research was conducted on **3D Digital Twin technology** for radioactive source localization. A real-time visualization system was developed by synchronizing live camera feedback with GPS coordinates and radiation data, enabling a three-dimensional representation of the monitored environment.

In addition, a complete backend framework was designed to integrate sensor communication, GPS telemetry, radiation processing, and visualization modules into a unified system. A **color grading algorithm**, inspired by night-vision imaging technology, was implemented to represent varying radiation intensity levels using distinct color scales on the live camera feed,

allowing intuitive identification of radiation hotspots. Furthermore, a **3D printable prototype model** of the RADMAP system was designed to represent the final hardware architecture and facilitate future prototype development.

The successful implementation and validation of the RADMAP prototype demonstrated the feasibility of real-time GPS-enabled radiation mapping using embedded computing and intelligent visualization techniques. The project received appreciation and funding approval from **Dr. Abhishek Yadav** and **Prof. (Dr.) Alpana Goel**, Director of AINST, for further enhancement toward next-generation radiation mapping, autonomous source localization, and advanced Digital Twin technologies. This internship provided valuable practical experience in embedded systems, IoT, geospatial analytics, radiation instrumentation, and research-oriented software development, while contributing to innovative solutions for nuclear safety and environmental monitoring.